SatSSTV Manual

Nov 8, 2026 · @f4lps

SatSSTV receives and sends pictures by radio in SSTV, KG-STV and DRM, tracks satellites and keeps your log. This manual describes version 0.56.0.

Installing

Download the file for your system, then open it. An update keeps your profiles, pictures and settings.

System	File	Installation
Windows	`SatSSTV-Setup-0.56.0.exe`	Open the file. It needs no administrator rights and also installs the Tracker F4LPS.
Linux 64-bit (Ubuntu 20.04, Debian 11, Mint 20 or newer)	`SatSSTV-0.56.0-x86_64.AppImage`	`chmod +x SatSSTV-*.AppImage`, then open the file.
Linux 64-bit, package	`satsstv_0.56.0_amd64.deb`	`sudo apt install ./satsstv_0.56.0_amd64.deb`. SatSSTV appears in the menu.
Mac (macOS 11 or newer)	`SatSSTV-0.56.0-macOS.dmg`	Drag SatSSTV into Applications, then follow the download page for the first opening.

- Windows and Linux: latest version
- Mac: latest version

On Linux, add your account to the `dialout` group to use the serial ports, then log in again: `sudo usermod -aG dialout $USER`.

On Mac, allow access to the microphone: without it, no reception is possible. The Mac version has no automatic update and does not include the Tracker F4LPS.

First start

Four settings are enough to receive your first picture. The menu is on the left of the window.

1. In **My station**, enter your callsign and your locator, for example JN18EU.
2. In **Radio & audio**, choose the interface that links the rig to the PC, or **None (no CAT)**.
3. Still in **Radio & audio**, choose the sound cards: input, headphones and transmit output.
4. Click **Save the profile**, at the bottom of the menu.

The size of the interface follows the window. To change it: **My station › Display**, or **Ctrl +**, **Ctrl −**, and **Ctrl 0** to go back to automatic.

Language

SatSSTV speaks French, English, German, Italian and Spanish. Choose in **My station › Display › Langue · Language**; it applies at once. By default, SatSSTV takes the language of Windows, or of the system on Linux and macOS.

The whole interface is translated: menus, buttons, help bubbles and messages. What you type, such as callsigns or the texts of the pictures, is never changed.

Connecting the rig

Choose the interface in **Radio & audio › Connection to the rig**. It reads and sets the frequency, and keys the PTT.

Interface	Use it when	Settings
None (no CAT)	The rig has no CAT cable	None. Frequency and mode are set on the rig.
Icom CI-V direct, or Win4Icom port	Icom connected by USB, or Win4Icom already open	Model, COM port, speed, CI-V address
Yaesu CAT direct	Yaesu connected by USB or CAT cable	Model, COM port, speed of the CAT RATE menu. FT-817/818/857/897 with the sound through the DATA jack: tick « DIG data mode instead of USB / LSB »
Kenwood / Elecraft CAT direct	Kenwood, Elecraft, or SDRuno	Model, COM port, speed
OmniRig	OmniRig is already in use (Windows only)	Rig 1 or Rig 2
Ham Radio Deluxe	HRD Rig Control is open and connected	IP and port 7809

Interface	Use it when	Settings
Hamlib / rigctld	rigctld runs with the right model	IP and port 4532
FLRig	FLRig is open and connected	IP and port 12345

After setting it, click **Read the frequency / test**. **Apply to the rig** really changes the frequency.

With Win4Icom, do not take the port Win4Icom already opened. Take the other end of the virtual pair of an auxiliary port, with the same speed.

Without CAT

Choose **None (no CAT)**. The PTT then goes through the VOX of the rig, or by hand.

- Copy the rig frequency into the box at the top of the workshops: it is the one that goes into the LOG.
- A click on an SSTV frequency tells which frequency and mode to set on the rig.
- The S-meter, the suggested report and the automatic Doppler are not available.

With a Digirig or a transistor interface, the PTT goes through an RTS or DTR line of the serial port, even without CAT: see **Digirig** below.

PTT choice

In the SSTV workshop, open **TX / PTT sources and settings**. Choose **PTT by CAT**, **PTT RTS / DTR (Digirig)** or **VOX / manual PTT**. This choice also applies to the DRM and Turbo workshops.

The **Before audio** (300 ms by default) and **After audio** (200 ms) delays are set in the same place.

Digirig

The Digirig links the rig to the PC with a single USB cable. It brings a sound card, « USB Audio CODEC », and a COM port whose RTS line keys the PTT.

1. Plug the Digirig audio cable into the data jack of the rig (DATA, PACKET or ACC depending on the model), with the cable made for that rig. If the rig has a serial CAT, also plug the Digirig CAT cable into its CAT jack.
2. In **Radio & audio › Connection to the rig**, choose the rig and the Digirig COM port when the CAT goes through it. Otherwise, choose **None (no CAT)**.

3. Tick **PTT by the RTS or DTR line of a serial port (Digirig…)**, choose the Digirig COM port, then **RTS**.
4. Choose **USB Audio CODEC** as audio input and output.
5. In the SSTV workshop, choose **PTT RTS / DTR (Digirig)**. Start with a low **TX audio level** with 🎵 **Tune 1750 Hz**, then raise it up to the ALC threshold.

When the CAT and the PTT share the Digirig port, SatSSTV keeps RTS at rest while reading the frequency: the rig does not go into transmit.

A rig whose CAT is not in the list (FT-1000 for example) works with **None (no CAT)** and the PTT through the Digirig; the frequency is then set on the rig. Some rigs only take the sound of their data jack in a data mode (PKT, DATA-USB): check the rig manual.

Sound

The COM port carries the CAT, not the sound. The sound goes through a sound card, chosen in **Radio & audio › Sound cards**. Without CAT, the transmitted sound triggers the VOX of the rig.

Setting	Choose
Input / reception	The USB Audio CODEC of the rig, or the output of a virtual audio cable fed by your remote rig software
Headphones / RX monitor	Your headphones or speakers
Output / SSTV transmission	The sound card connected to the audio input of the rig

Click **Start RX listening** to hear the rig and see the level. The level is right if the bar does not turn red. **Refresh audio and COM ports** reads the cards again after plugging one in.

With an SDR, you need a virtual audio input. On Linux, the virtual cable is replaced by a « monitor » output of PulseAudio or PipeWire. On Mac, use BlackHole for example.

Receiving an SSTV picture

Leave **Auto · VIS** and click ▶ **Start RX**: SatSSTV recognises the mode of each picture by itself.

1. In the SSTV workshop, open the **SSTV frequencies** tab and click a frequency. The rig goes there with the mode of the band.
2. Check that the chosen mode is **Auto · VIS**.

3. Click ▶ **Start RX**. The spectrum and the waterfall show the received signal.
4. The picture is drawn line by line, then goes into the **History SSTV** if **Auto save** is ticked.

Reception goes on while you look at the other pages. The 16 modes received and sent are Martin 1 and 2, Scottie 1, 2 and DX, Robot 36 and 72, PD50 to PD290, and Wraase SC2-60 and SC2-180.

In the settings of the SSTV workshop, **Start RX listening when the program opens** puts SatSSTV in reception as soon as it opens: every picture announced by its VIS is decoded without a click.

Usual frequency	Rig mode	Use
14.230 MHz	USB	Main worldwide frequency
7.165 MHz	LSB	Europe
3.735 MHz	LSB	Europe
21.340 MHz	USB	Worldwide
28.680 MHz	USB	Worldwide
437.550 MHz	FM	ISS, ARISS pictures in Robot 36

The **SSTV frequencies** tab gives the full list, from 80 m to 70 cm.

- **Picture already started:** choose its mode, then click **Start without VIS**.
- **Slanted picture:** keep **Auto slant** ticked. The **Sync** tab corrects the sound card clock and retouches a received picture.
- **WAV recording:** the **Replay a reception** tab decodes a file.

Force reception

If the start of a picture was missed (VIS not received), open ⚡ **Force reception** and choose the mode of the picture. Decoding starts at once, in the middle of the picture, then the automatic mode takes over again for the next pictures.

Reception report on the picture

Each received picture keeps its information: UTC time, mode, frequency, satellite, orbit and elevation, and the callsign and RSV report if they were entered in the LOG during reception. They show in the gallery and in the viewing window.

The copy put in **My reports folder** carries this information in a strip under the picture, and its name follows the form `SSTV_YYYYMMDD-HHMMSSUTC_MODE`. The original picture stays untouched in the History. **Also save each received picture as .jpg** adds a JPG copy.

Reception quality

Noise reduction, next to **Auto slant** and ticked by default, smooths the noise at the end of the picture, the more the stronger it is, without blurring the edges. A clean received picture is not changed.

The colours of the PD and Robot modes follow the scale of MMSSTV, the ISS and Robot36. If a correspondent sends with QSSTV, tick **QSSTV PD and Robot colours** in the settings of the SSTV workshop; Martin, Scottie and SC2 are not concerned.

SatSSTV corrects the tuning of the rig from the syncs (AFC): a rig slightly off frequency, or drifting during the picture, keeps the right colours, even if the VIS was not received.

If the sound card drops out a moment (USB cable, driver), listening resumes by itself and the picture goes on; the message « Sound cut: restarting the sound card… » shows meanwhile.

The last 20 pictures

The received picture window is also a history, without opening a gallery: handy to give a report. The arrows are on the left of the picture.

Button	Action
◀◀	Back to the last received picture
◀	More recent picture
▶	Older picture
▶▶	Goes to the oldest of the list

The list shows the last 20 pictures of the History SSTV. When a new picture arrives, the oldest leaves the list but stays in the History. **Auto save** must be ticked for a picture to enter the list.

A right click on the picture offers to show it large, to reply with it, to put it into the picture to send, to copy it, to make it a favourite or to open its folder. **Remove from this list** only takes it out of the list, without deleting anything in the History.

If a picture arrives while you look at an older one, the screen does not change. A green line says so, and ◀◀ brings back the current picture.

Sending an SSTV picture

Compose the picture, choose the mode, prepare the signal, then transmit. A yellow line follows the progress on the picture.

1. In **Picture to send**, start from a **Photo** or a **Template**, then add **Cliparts** or **Text**. The elements are moved directly on the picture. After a click on the picture, the arrow keys move the chosen element by one pixel, by 10 with Shift.
2. Just under the picture, choose the **Mode of the picture to send**.
3. Click **TX · Prepare the picture**.
4. Click ▶ **TX · Transmit the picture**.
5. At the end, the program goes back to reception by itself. With manual PTT, release the PTT of the rig.

To cut the transmission: ■ **STOP TX / PTT OFF**. Changing the mode cancels the prepared signal: prepare it again.

Layout of the SSTV workshop

On a wide enough screen, the received picture and the picture to send are side by side, at the same height and size; the sending settings (photo, mode, FSK-ID and CW, TX buttons) are in a column on the right. In a narrow window, the page goes back to two columns.

Hide waterfall and LOG reduces the page to the pictures: the Call, Name, Report, Locator fields and **Log the QSO…** then stay on the levels line, right of the TX level. **Show waterfall and LOG** brings back the full LOG.

Moving the mouse over the received picture shows it large near the pointer; the right click keeps its usual menu.

Saving your layout as a template

💾 **Save as template…**, at the top of the sending settings column (or a right click on the picture to send), keeps the texts, frames, picture of the OM and cliparts as a new template. A window asks for its name, its group and whether to keep the background photo. The template appears first in the template gallery.

In the list of groups, ➕ **New group…** creates a group of your own; it then has its own tab in the template gallery. Next to it, 🖼️ **Save the picture as PNG…** saves the picture to send as it will go out, in a PNG file.

The **SatSSTV banner** template copies the MMSSTV banner: a strip at the top with your callsign and « SatSSTV Ver … ». Click the strip to set its colour and the colour of its text.

Setting the level

🎵 **Tune 1750 Hz** sends a steady tone, PTT included, until ■ **STOP TUNE**. It stops by itself after 5 minutes.

Set the **TX audio level** to avoid ALC saturation. This setting also applies to DRM and Turbo.

The **TX level** slider is also under the spectrum, next to **RX gain**, and ⚙ **TX** opens the transmit settings. The level can be changed during transmission too: SSTV, Tune, DRM and Turbo. The same RX gain · Auto · TX level · ⚙ TX line is in the DRM and Turbo workshops, above the waterfall. There, the RX gain also amplifies the sound received in DRM.

Listen on headphones without PTT plays the signal without transmitting.

Callsign after the picture

These two options are in the sending settings column, right of **Mode of the picture to send**.

- **Callsign in FSK-ID after the picture:** your callsign goes out in FSK. MMSSTV, YONIQ and SatSSTV read it and put it in the LOG.
- **Callsign in CW at the end:** a text in Morse. Empty, it sends DE and your callsign. Set its speed (5 to 40 words/min) and its tone (300 to 2700 Hz).

Variables

The **{ } Variables** button, next to **+ Text** in Picture to send, lists all the variables. A click adds the variable to the chosen text, or creates a new text. The values are filled in **Contact and variables**, in the Templates tab.

Variable	Becomes
`{MYCALL}`	my callsign
`{LOCATOR}`	my locator
`{CALL}`	callsign of the correspondent
`{CALL_DE_MYCALL}`	the correspondent's callsign then mine: « F1ABC de F4LPS »
`{RST}`	report
`{NAME}`	first name of the correspondent
`{QTH}`	QTH of the correspondent
`{UTC}`	UTC date and time when the signal is prepared

Variable	Becomes
`{TIME_UTC}`	UTC time only
`{TIME_LOCAL}`	local time only
`{73}`	« 73 de » and your callsign
`{SAT}`, `{ORBIT}`, `{ELEV}`	satellite, orbit number, elevation
`{SATSSTV}`	« SatSSTV Ver » and the version
`{SATSSTV_UTC}`	the same, followed by the UTC date and time of sending

KG-STV (test version)

KG-STV, by JJ0OBZ, is a digital SSTV widely used on QO-100: the 320 × 240 picture goes out in 300 blocks, each one checked on arrival. This mode is a **test version** in SatSSTV 0.56.0: your feedback is welcome.

Receiving: nothing to set. KG-STV is listened to at the same time as SSTV in the SSTV workshop. The picture fills block by block (« KG-STV from …: 150/300 blocks »), then goes into the mosaic, and the received callsign fills the Call field like an FSK-ID. **KG-STV only**, at the bottom of the list of reception modes, listens to KG-STV alone.

Sending:

1. Under the picture to send, choose the mode **KG-STV (digital, QO-100)**.
2. Choose **MSK** (the safest) or **4L-FSK** (faster, for a strong signal), **Error correction** (the picture gets through with a weaker signal, but sending takes longer) and the **Compression** (1.00 usually).
3. Click **TX · Prepare the picture**, then **▶ TX · Transmit the picture**.

The picture is cropped to 4:3 and reduced to 320 × 240, and your callsign goes out with it. Sending takes from about 1 min 30 (4L-FSK) to 3 min 30 (MSK with correction). The rig stays in USB, as for SSTV.

Missing blocks (BSR): after an incomplete picture, **Ask for the missing blocks (BSR)** prepares the request, sent with ▶ TX; the blocks sent again complete the picture in the mosaic. When a correspondent asks for blocks of your last picture, **Prepare the answer** appears under the reception settings.

Templates, galleries and History

Received pictures go into the History of their mode: SSTV, DRM and Turbo stay apart. Each workshop also has its templates.

Right click on a picture	Effect
📋 **Copy**	Paste the file with Ctrl+V into a folder
🖼️ **Copy the picture**	Paste the picture into Discord, Word, etc.
📁 **Copy to another folder…**	Choose the destination folder

- **Several pictures:** tick **Multiple selection**, then copy or delete. You can also drag the pictures to a folder.
- **My reports folder:** at the top of each History, choose your folder. It is remembered, and each received picture can be copied there automatically.
- **Separate window:** 🪟 opens a gallery in its own window, to drag onto a second screen.
- **Templates:** the groups are CQ, QSO reply, Info and End of QSO. You can create a template or save a copy.
- **Import from MMSSTV or YONIQ:** Templates tab, 📥 Import MMSSTV / YONIQ… button. The window opens on the MMSSTV Stock folder, and Ctrl+A takes all the .mtm files. YONIQ uses the same files. The originals are not changed, and a second import creates no duplicates.
- **Mosaic:** ▦ **Mosaic** puts several received pictures into one, 2 side by side, 2 × 2, 3 × 2 or 3 × 3.
- **Right click on a text:** replaces this text with a received picture or a clipart, or removes it alone with **Delete this text**, keeping the rest of the template. A new text starts empty: no « Text » word is written on the picture.
- **Photos:** an added photo is brought to SSTV proportions, 640 × 512 or 320 × 256. Choose at the top of the gallery: cropped, whole with black bands, or unchanged.
- **Photo settings:** under the picture to send, 🔆 Photo settings sets brightness, contrast, colours and sharpness, and rotates or flips the photo. Only the background photo changes, not the texts. The original photo stays untouched, and another photo starts from neutral settings.
- **Zoom and cropping:** in the photo settings, the Zoom, horizontal and vertical sliders enlarge the photo and choose the part kept in the TX frame.
- **Naming a template:** Name field in the template editor; a right click on a template of the grid offers ✏ **Rename…**.

- **Photos folder:** the gallery of photos to send shows the folder where they are, and the 📂 button opens it.
- **Room for an information bar:** in the photo settings, the top and bottom sliders reduce the photo, without distorting it, to leave room for a bar (callsign, report), as in MMSSTV. A menu chooses how the photo fills the room: reduced with black bands on the sides, or over the whole width, reduced in height or with the top and bottom cut. The setting is kept for the next photos.
- **Added photos: automatic:** a photo close to the SSTV format is cropped; a portrait or very long photo is kept whole, with black bands.

What changes when importing from MMSSTV

Texts, gradient strips, frames for the received picture, fonts, bold, shadows and outlines are kept, and the templates stay editable. `%m`, `%c` and `%r` become `{MYCALL}`, `{CALL}` and `{RST}`. A few effects differ:

- Text in perspective is drawn straight.
- The stacked 3D effect becomes a simple shadow.
- The gradient of the letters goes from top to bottom.
- Only Comic Sans, Arial Black, Impact and the fixed-width fonts are kept. The others become the standard font.
- `MMSSTV %v` becomes `SatSSTV` followed by its version.

LOG and QSO

The LOG frame, under the waterfall, keeps the record of the current correspondent: callsign, name, report, QTH, locator and country.

- **Received callsign:** an FSK-ID fills the **Call** field, whether it arrives before or after the picture. A line under the LOG tells what happened to the last FSK-ID received.
- **Suggested report:** it is computed for each received picture, in RSV format. Example: 575 for a complete picture received at S7. Without S-meter, the report is typed by hand.
- **Look up the OM:** fills the first name, QTH, locator and country from the Tracker F4LPS, QRZ.com, HamQTH, HamDB, Callook or public pages.
- **Log the QSO…:** records the contact with its date, time and frequency.
- **Remember:** adds the OM to the list of the last 100 correspondents.
- **New QSO:** clears the record and keeps the list.

Logbooks and services

In **My station**, ⚙ **Log settings** configures the ADIF file, HRD, N1MM+, Log4OM, eQSL, ClubLog, WaveLog, QRZ.com and LoTW.

The QRZ.com (XML subscription) and HamQTH (free) accounts are also entered in **My station**. The **Take the logins…** button gets them from the Tracker, CW Terminal or MultiDigi.

Tracker F4LPS, in the menu, opens the map shared with MultiDigi and CW Terminal. Received callsigns can be sent to it.

DRM workshop

The DRM workshop exchanges digital pictures and files with EasyPal and QSSTV. The file arrives without loss, even if segments must be asked for again.

Set the rig to USB, for example on 14.233, 7.173 or 3.733 MHz, with a wide passband and the ALC at zero.

Receiving

Click ▶ **Start DRM RX**. The mode, the QAM and the bandwidth are recognised by themselves.

- The grid shows the received segments. **Ask for the missing ones (BSR)** prepares a request, sent with the TX button.
- The MER tells the quality: at 20 dB and more, reception is comfortable. Under 12 dB in QAM 16, go to QAM 4 or mode E.
- A WAV file is decoded with **Browse…**, then **Decode the WAV**.

Sending

1. Choose a photo, the picture composed in the SSTV workshop, a picture of the PC, or **Send a file…**.
2. Choose the mode, the QAM, the protection, the interleaving and the Reed-Solomon.
3. Click **Prepare the DRM signal**, then ▶ **TX · Transmit in DRM**.

The segment grid shows what goes out: grey to send, green sent, yellow on the air. The level, the PTT and the delays are those of SSTV transmission.

- **Voice message:** record 30 s at most. The voice is compressed in Opus, about 0.75 kB per second at 6 kbit/s.
- **Large file:** tick sending in several parts, up to 5 MB in all. Splitting is required above 400 kB.

- **Text only:** 📝 **Send to the waterfall** writes a message visible in the waterfall, without a picture.

In DRM, the power swings and the average is about half that of the tune: this is normal. Set the watts with the power knob of the rig.

Sound going through remote rig software is often compressed and lowers the MER. Then prefer QAM 4 and a high protection.

Hybrid DRM

In hybrid mode, the picture goes through an FTP server and the radio only sends where to fetch it. The transmission lasts about 25 seconds whatever the size of the picture, but both stations need the Internet. The format is that of EasyPal and QSSTV.

To send:

1. Tick 🌐 **Send in hybrid mode**, under the transmit settings.
2. Enter the server, the user, the password and the folder, then click **Test the server**.
3. Click **Prepare the DRM signal**: the picture goes to the server.
4. Click ▶ **TX · Transmit in DRM**.

The server, the user and the password go out on the air, readable by everyone. Use an FTP account kept for the pictures, never a personal account.

To receive: tick 🌐 **Receive hybrid pictures**, off by default. The server is chosen by the sending station: SatSSTV refuses private addresses and files over 8 MB. It neither stores nor shows the received password.

HTTPS hybrid with a token

Between two SatSSTV stations (version 0.54.0 or newer), hybrid mode can use HTTPS instead of FTP. The picture is put on your web hosting with a **secret token, which never goes on the air**. The radio only sends the address to read the picture: nobody can put or delete a file on your server. EasyPal and QSSTV cannot read this announcement: keep FTP for them.

You need web hosting with PHP and HTTPS. The free alwaysdata offer (1 GB) is enough.

Preparing the server (once):

1. Create an account on alwaysdata.com, **Free** offer. The chosen name gives the site address, for example `https://f4xxx.alwaysdata.net`. The password is also used for FTP.
2. In SatSSTV, DRM workshop, tick 🌐 **Send in hybrid mode** and choose **HTTPS with token**.

3. In **Upload script**, enter `https://f4xxx.alwaysdata.net/sstv/satsstv-hybride.php`.
4. Click 🎲 **New token**, then 💾 **Save the server script…**: the file `satsstv-hybride.php` contains your token.
5. In alwaysdata, **Remote access** › **FTP** gives the host (`ftp-f4xxx.alwaysdata.net`) and the user. With FileZilla Client, connect with « explicit FTP over TLS », create the `sstv` folder in `www`, and put the script there.
6. Click **Test the server**: SatSSTV must answer that the script accepts the token.

Then sending works as with FTP: **Prepare the DRM signal** puts the picture on the server, then ▶ **TX · Transmit in DRM**. The script keeps the pictures in an `images` folder, refuses files over 8 MB and unknown types, and deletes them after 3 days.

If you click **New token** again, save the script again and put it back on the server. Never share the script file: it contains your token.

Turbo workshop

Turbo mode only works between two SatSSTV (version 0.45.0 or newer). EasyPal, QSSTV and HamDRM do not decode it.

It takes the same 2.5 kHz as DRM and is 1.3 to 1.7 times faster for the same picture. Everything works as in the DRM workshop: pictures, files, voice messages, sending in parts and BSR.

Code	Use
1/2	The most robust
2/3	In between
3/4	Balanced, the default
5/6	The fastest, needs a good signal

The correspondent must put his Turbo workshop in reception. This mode was tested in simulation: your on-air reports are welcome.

Satellites and Doppler

The **Satellites & Doppler** page computes the passes over your locator, follows the satellite on the globe and can receive a pass without you. It needs an orbit catalogue less than 14 days old.

The catalogue is **updated automatically** every day when **Update the orbits automatically** is ticked. Pasting a catalogue by hand unticks the box, to keep it. The **Other satellites** button shows on the map the SSTV satellites and those of the armed passes, in green when they are visible.

The two lists, **SSTV satellites** and **Amateur radio satellites**, are kept after their download: going back to one is immediate. CelesTrak accepts only one download per list every 2 hours; a greyed button tells in how many minutes.

1. Enter your locator in **My station**.
2. In **Orbital catalogue**, load the orbits, or paste a TLE or OMM/JSON catalogue.
3. Choose the **Tracked satellite**: the list is in alphabetical order.
4. In **Doppler correction (reception)**, enter the nominal frequency, or click **ISS SSTV 437.550 MHz**.

Armed passes

An armed pass sets the rig before AOS, corrects the Doppler and starts reception. During that pass, transmission and PTT are blocked.

Add one or more satellites (the list is in alphabetical order), tick **above the threshold** or arm a pass, then leave the program open. **Stop the armed pass** interrupts it. The sound of the passes can be recorded to decode them again.

Without CAT, the program shows the frequency to set and the automatic Doppler stays inactive.

Alerts

My station › Pass alerts sets the delay before AOS (1 to 30 minutes) and the minimum elevation. Alerts for armed passes can also go out as system notifications, on Discord or on Telegram.

Weather workshop

The Weather workshop decodes the pictures of the Russian weather satellites **Meteor-M2 3** and **Meteor-M2 4** (LRPT, 137.900 MHz, or 137.100 MHz as backup frequency). Decoding uses an IQ recording made with an SDR dongle during a pass: SatSSTV does not drive the dongle itself.

Equipment

- An **RTL-SDR** dongle (V3 or V4) and a 137 MHz antenna: **QFH**, or **V-dipole** (2 elements of 53.4 cm opened at 120°, flat, tip pointing north or south).

- An antenna with a clear view of the sky, a coaxial cable as short as possible. A 137 MHz LNA preamplifier helps a lot; it can be powered by the bias-T of the dongle.
- A transceiver (IC-705, IC-7300…) is not suitable: its filter is much narrower than the 120 kHz of the LRPT signal.

Next passes

Click **Download the orbits (CelesTrak)**: the table gives the passes of the next 48 hours (local start time, maximum elevation), computed from the locator of **My station**. The orbits reload at most every 2 hours. Prefer a pass above 30° maximum elevation: the higher, the better the picture.

Recording a pass with SDR++

Guide: record a pass with SDR++ and an RTL-SDR dongle, unfolded in the page, gives these settings:

1. Source **RTL-SDR**, frequency **137.900 MHz** (137.100 MHz if the satellite switched to the backup frequency).
2. Sample rate **1.024 MHz** (about 4 GB for 15 minutes) or **250 kHz** (about 1 GB).
3. Manual gain around 35–40 dB, AGC off. Tick **Bias-T** only if a preamplifier is powered through the cable, and **IQ correction** to remove the line at the centre of the spectrum.
4. In the Recorder module: mode **Baseband** (not Audio), format **WAV**, samples **Int16**.
5. Start recording when the satellite rises and stop when it disappears. The signal is a flat hump of about 120 kHz drifting slowly: that is the Doppler, nothing to correct by hand.

SDR# (Baseband recorder, 16-bit WAV) and SatDump (baseband recording in WAV) also give readable files.

Decoding

1. In **Decode a recording**, click **Browse…** and choose the .wav file.
2. Leave **Modulation** on **Auto (OQPSK then QPSK)** and click **Decode**. **Stop** interrupts a long file.
3. The decoder finds the carrier by itself and follows the Doppler. The status line gives the carrier offset, the code error rate, the number of valid frames, picture packets and Reed-Solomon corrections.

The pictures are saved in the **Meteo** folder of the data folder; **Open the pictures folder** opens it. Depending on the channels received, SatSSTV produces:

Picture	Content
Natural colours 221 (day)	Channels 2, 2 and 1: the closest to the eye, by day
Composite 321	Channels 3, 2 and 1: snow and ice stand out from the clouds
Infrared (clouds in white)	Channel 5 or 6, inverted: readable at night too
One channel	Each received channel, in black and white (visible, near infrared, thermal infrared)

Click a thumbnail to show it large, **Fit to width** or at **Real size (1568 px)**.

If no picture comes out, the message gives the number of valid frames. Zero frames: the recording is not an IQ file centred on 137.9 or 137.1 MHz, or the signal is too weak (antenna, gain, elevation). The interleaved 80k mode is not supported yet.

For an SDR, **Radio & audio** offers the SDRplay · SDRuno and SDR++ presets. Tick **Receiver only (SDR)**: transmission and PTT are then disabled.

Troubleshooting

Most problems come from the chosen COM port or sound card. Check them first in **Radio & audio**.

Problem	Likely cause	Solution
« Read the frequency » fails	Wrong port, wrong speed, or port already in use	Check the COM port and the speed of the rig menu. Close the other software using that port.
Win4Icom: no connection	Auxiliary port off, or different speed	Enable the auxiliary port in Win4Icom and set the same speed in SatSSTV.
Nothing on the waterfall	Wrong audio input	Choose the input of the rig in **Sound cards**, then ▶ **Start RX**.
The RX level is red	Input saturated	Lower the **RX gain** or the output level of the rig.
The VOX does not trigger	Level too low or VOX off	Turn on the VOX of the rig, then raise the **TX audio level** with **Tune**.

Problem	Likely cause	Solution
The rig transmits but sends no modulation	Data jack not used in this rig mode, or wrong output	Check the TX output in **Sound cards**; on some rigs, use a data mode (PKT, DATA-USB) or the microphone input.
The received picture is slanted	Sound card clock	Keep **Auto slant**, then calibrate in the **Sync** tab.
Shifted lines at the bottom of the picture	Sending station older than 0.49.0, or poor sound	Update SatSSTV on both sides.
A few shifted lines in the middle of the picture	Sound cut a moment, when the PC is very busy	Look at « sound cuts: N » under the spectrum. Avoid heavy tasks during reception, and send a screenshot of the picture with that number.
The FSK-ID does not show in MMSSTV	Same callsign on both sides, or QSO button pressed with a callsign already entered	Try another callsign. Release QSO or clear the field.
The report stays at 595	S-meter not read	Tick **Follow the rig**, or type the report by hand.
DRM: missing segments	QSB or weak signal	Click **Ask for the missing ones (BSR)**, then go to QAM 4.
Turbo: nothing received	The correspondent is not in Turbo	Check that both stations use the Turbo workshop.
KG-STV: blocks missing	Weak signal or interference	Click **Ask for the missing blocks (BSR)**; for weak signals, send in MSK with error correction.
Mac: « application is damaged »	Application not signed by Apple	Run `xattr -cr /Applications/SatSSTV.app` in the Terminal.
A received picture is missing from the last 20	Auto save off, or picture removed from the list	Tick Auto save. A picture removed from the list stays in the History SSTV.
Mac: USB AUDIO CODEC missing from the output	Old SatSSTV version, or card badly described by macOS	Install the latest version, click Refresh audio and COM ports. If the card is still missing, send the text of the 📋 Copy the audio diagnostic button.

Where your files are

Everything is in a `SatSSTV` folder, kept through updates.

System	Location
Windows	`%LOCALAPPDATA%\SatSSTV`
Linux	`~/.local/share/SatSSTV`
Mac	`~/Library/Application Support/SatSSTV`

The received pictures are in `SSTV/history`, `DRM/history` and `Turbo/history`. Do not delete the `.json` files next to the SSTV pictures: they are their records (date, mode, favourite).

To change the location, go to **My station › Data folder**, then **Choose another folder…**. The data is copied and used at the next start, and the old folder is kept.

Two stations on one PC: install SatSSTV in two folders and create, next to `satsstv.exe`, an empty folder named `sstv-data`, before the first start. Each copy keeps its own settings (COM port, sound cards, rig) and its pictures there.